

In this case, a NMI is generated. The user has to read ECCD and ECCD2 to see which part of the 128-bits data has error detection (either 63:0, 127:64 or both).

When an ECC error is detected, the address of the failing the two times double word is saved into ADDR_ECC[18:0] in FLASH_ECCR. ADDR_ECC[18:0] contains an address of a two times double word.

The ADDR_ECC[3:0] are always cleared. BK_ECC is not used in this mode.

When ECCC/ECCC2 or ECCD/ECCD2 is/are set, if a new ECC error occurs, the ADDR_ECC is not updated. The FLASH_ECCR is updated only if the ECC flags (ECCC/ECCC2/ECCD/ECCD2) are cleared.

Note: *For a virgin data: 0xFF FFFF FFFF FFFF FFFF, one error is detected and corrected but two errors detection is not supported.*

When an ECC error is reported, a new read at the failing address may not generate an ECC error if the data is still present in the current buffer, even if ECCC and ECCD are cleared.

6.3.3 Read access latency

To correctly read data from Flash memory, the number of wait states (LATENCY) must be correctly programmed in the [FLASH registers](#) according to the frequency of the CPU clock (HCLK) and the internal voltage range of the device V_{CORE} . Refer to [Section 8.2.5: Dynamic voltage scaling management](#). [Table 32](#) shows the correspondence between wait states and CPU clock frequency.

Table 32. Number of wait states according to CPU clock (HCLK) frequency

Wait states (WS) (Latency)	HCLK (MHz)		
	V_{CORE} Range 0	V_{CORE} Range 1	V_{CORE} Range 2
0 WS (1 CPU cycles)	≤20	≤20	≤8
1 WS (2 CPU cycles)	≤40	≤40	≤16
2 WS (3 CPU cycles)	≤60	≤60	≤26
3 WS (4 CPU cycles)	≤80	≤80	-
4 WS (5 CPU cycles)	≤100	-	-
5 WS (6 CPU cycles)	≤110	-	-

After reset, the CPU clock frequency is 4 MHz and 0 wait state (WS) is configured in the FLASH_ACR register.

When changing the CPU frequency, the following software sequences must be applied in order to tune the number of wait states needed to access the Flash memory: